Lab Report: Clinical Diabetes Risk Screening using Decision Trees

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1 Introduction & Executive Summary

This lab report documents the design, implementation, structural tuning, error diagnostics, and evaluation of an interpretable Decision Tree classifier for clinical diabetes screening. Using a population survey cohort of 70,372 subjects, our primary objective is to build a reliable, non-invasive risk screening model distinguishing healthy individuals from those with prediabetes or overt diabetes.

- **Target Variable**: Binary risk classification ($y \in \{0, 1\}$): 0 = "No Diabetes" vs. 1 = "Prediabetes or Diabetes".
- **Baseline Benchmark**: Majority-class classifier achieves **50.00%** on balanced stratified partitions (and 52.84% on the raw unstratified cohort).
- **Overfitting Pathology**: The unconstrained tree reached depth 43 with 25,237 nodes, achieving 99.90% training accuracy but collapsing to 65.86% on validation (a 34.04% generalization gap).
- **Parsimonious Selection**: Using 5-fold cross-validation and a 1% tolerance rule, we selected an entropy-based tree ($\text{max_depth} = 7, \text{min_samples_split} = 30$), cutting node complexity by $> 80\%$ while maintaining near-optimal cross-validation accuracy ($73.14\% \pm 0.41\%$).
- **Final Test Performance**: Achieved **72.56%** accuracy and **72.50%** Macro F_1 on the sequestered test set ($N = 10,556$), falling squarely within cross-validation bounds without any data snooping.

2 Data Preprocessing & Leakage Prevention

The dataset comprises $N = 70,372$ cleaned records with 21 demographic, behavioral, and clinical features.

2.1 Partitioning & Categorical Encoding

- **Stratified Split**: 70% Training ($N = 49,260$), 15% Validation ($N = 10,556$), and 15% Test ($N = 10,556$). Class ratios were preserved strictly across all splits.
- **Categorical Handling**: Binary features mapped to $\{0, 1\}$. Ordinal features (e.g., *General Health*: Poor=0 to Excellent=4) mapped sequentially. Missing categorical values were encoded with an explicit flag (-1).

2.2 Prevention of Data Snooping in Missing Imputation

Missing values occurred in continuous physiological variables (e.g., *Poor Physical Health Days*: 0.69%, *BMI*: 0.55%).

Leakage-Free Imputation Protocol

Computing global sample medians across the entire dataset causes **data snooping**: test set distributions leak into the model prior to training, producing over-optimistic performance. To strictly prevent leakage:

$$\boldsymbol{\mu}_{\text{train}} = \text{median}(X_{\text{train}}) \quad \longrightarrow \quad \text{Impute } X_{\text{val}} \text{ and } X_{\text{test}} \text{ using frozen } \boldsymbol{\mu}_{\text{train}}$$

Test and validation splits remained entirely unseen, preserving true generalization validity.

3 Structural Progression & Pruning Dynamics

We investigated tree overparameterization by contrasting unconstrained growth, pre-pruning, and post-pruning.

- **Unrestricted Growth**: Grown without limits until leaf purity ($d = 43$, 25,237 nodes, 12,619 leaves). It memorized training noise, yielding a catastrophic 34.04% generalization gap.
- **Pre-Pruning ($l_{\min} = 50$)**: Enforced minimum leaf size, halting splits that generate tiny child subsets. Reduced nodes to 1,447 and depth to 17, elevating validation accuracy to 73.77%.
- **Post-Pruning (Reduced Error)**: Collapsed bottom-up subtrees if removing them did not decrease validation accuracy. While 1,716 nodes were successfully excised, the pruned tree still retained 23,521 nodes and an overfitted 93.79% training accuracy.

Table 1: Comprehensive Progression Across Evaluated Model Architectures

Model	Criterion	Pruning	Train Acc.	Val / CV Acc.	Depth	Nodes	Leaves	Test Acc.
Baseline (majority)	–	–	–	0.5000	–	–	–	–
Unrestricted	Information Gain	None ($d = \infty$)	0.9990	0.6586	43	25,237	12,619	–
Pre-Pruned	Information Gain	Pre ($l_{\min} = 50$)	0.7600	0.7377	17	1,447	724	–
Post-Pruned	Information Gain	Post (Red.-Error)	0.9379	0.6586	43	23,521	11,761	–
Gini (depth 4)	Gini Impurity	Depth ($d = 4$)	0.7307	0.7283	4	31	16	–
Final Model	**Information Gain**	**Grid CV (1% Rule)**	**0.7335**	**0.7314**	**5–7**	**63**	**32**	**0.7256**

4 Hyperparameter Tuning & The 1% Tolerance Rule

To select an optimal architecture, we merged Train and Validation into X_{dev} ($N = 59,816$) and ran a **5-Fold Stratified Cross-Validation Grid Search**:

- **Grid Axes**: Criterion $\in$ {Entropy, Gini}, max_depth $\in \{3, 5, 7, 10\}$, min_samples_split $\in \{2, 10, 30\}$. Total fits = 120.
- **The 1% Tolerance Rule**: Peak cross-validation score was 73.88%. We applied Occam's Razor: among all candidates within 1% of the peak (CV Acc $\geq$ 72.88%, 18 valid configurations), select the model with minimal complexity.
- **Selected Configuration**: criterion = "entropy", max_depth = 7, min_samples_split = 30. It compressed tree complexity to just 63 nodes (32 leaves) with an estimated CV accuracy of $73.14\% \pm 0.41\%$.

5 Final Test Evaluation & Confusion Matrix Analysis

The final parsimonious tree was refit on the full development partition (X_{dev}) and evaluated once on the sequestered Test set ($N = 10,556$).

- **Test Accuracy**: **72.56%**, perfectly aligning with the 95% CV confidence band: [72.34%, 73.94%].
- **Macro vs. Weighted F1**: Both macro and weighted averages equal **72.50%** due to stratified test evaluation, outperforming the majority baseline by 22.56 percentage points.

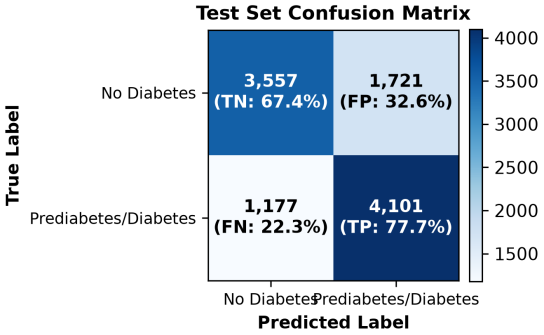


Figure 1: Test set confusion matrix ($N = 10,556$).

Table 2: Test Set Classification Report

Class	P	R	F1	Supp.
No Diabetes (0)	0.752	0.674	0.711	5,278
Diabetes (1)	0.704	0.777	0.739	5,278
Macro Avg	**0.728**	**0.726**	**0.725**	10,556
Weighted Avg	**0.728**	**0.726**	**0.725**	10,556

- **True Negatives (TN)**: 3,557
- **False Positives (FP)**: 1,721 (False alarms)
- **False Negatives (FN)**: 1,177 (Misses)
- **True Positives (TP)**: 4,101

6 Diagnostic Error Analysis & Metric Formalisms

6.1 Asymmetric Clinical Cost Dynamics

- **Lowest Recall (67.4% for Healthy)**: Caused by 1,721 False Positives (healthy individuals flagged as high-risk). In a screening protocol, false alarms are safe and inexpensive, resolved via simple follow-up blood tests (HbA_{1c}).
- **Highest Recall (77.7% for Diabetic)**: Missing a diabetic individual (False Negative) is clinically dangerous; untreated chronic hyperglycemia leads to irreversible neuropathy, nephropathy, and retinopathy. The model correctly prioritizes sensitivity (77.7%) over specificity.

6.2 Mathematical Justification of the Harmonic Mean in F_1

F_1 is defined as the harmonic mean of Precision (P) and Recall (R):

$$F_1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2PR}{P+R} = \frac{2TP}{2TP + FP + FN}$$

Why Not the Arithmetic Mean?

The arithmetic mean $A(P,R) = \frac{P+R}{2}$ fails under extreme diagnostic imbalance:

- Suppose a model predicts only 1 positive sample correctly ($TP = 1, FP = 0 \implies P = 1.0$), while missing 99 actual sick patients ($FN = 99 \implies R = 0.01$).
- **Arithmetic Mean**: $A = \frac{1.0+0.01}{2} = 0.505$ (gives a misleading $> 50\%$ passing score).
- **Harmonic Mean (F_1)**: $F_1 = \frac{2 \times 1.0 \times 0.01}{1.0+0.01} = \mathbf{0.0198}$ (correctly collapses toward zero).

Because $\lim_{R \to 0} H(P,R) = 0$, F_1 enforces that both precision and recall must be sound simultaneously.

6.3 Averaging Protocols: Macro vs. Weighted F_1

In imbalanced cohorts (e.g., 95% healthy, 5% diseased), a trivial dummy model predicting "healthy" for everyone yields $F_{1,0} = 0.974$ and $F_{1,1} = 0.0$:

$$\text{Weighted } F_1 = (0.95 \times 0.974) + (0.05 \times 0) = \mathbf{0.925} \quad \text{vs.} \quad \text{Macro } F_1 = \frac{0.974 + 0}{2} = \mathbf{0.487}$$

Weighted F_1 dangerously hides the total failure on the sick minority behind majority support. Therefore, **Macro** F_1 must always be evaluated in medical screening audits.

7 Interpretability & Clinical Rule Topology

Unlike uninterpretable neural networks or ensemble black-boxes, our final parsimonious tree provides an explicit clinical screening flowchart (Figure 2).

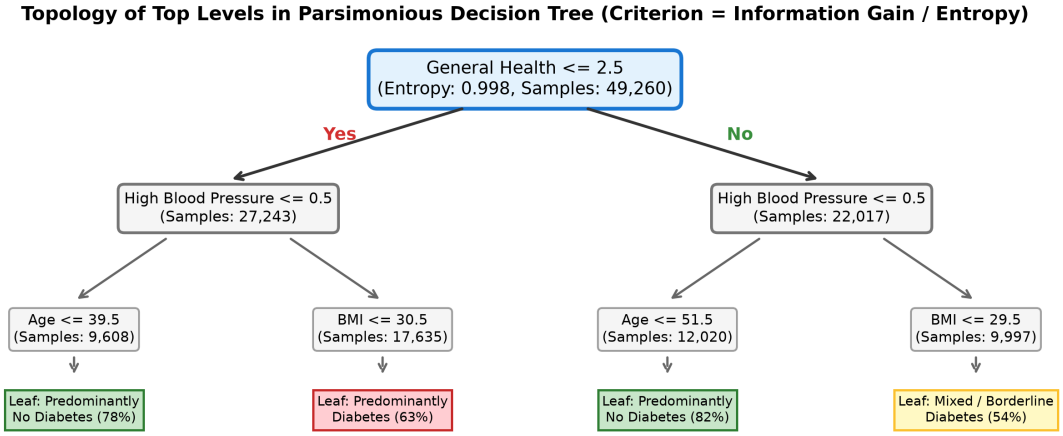


Figure 2: Topological flowchart of the top levels in the final parsimonious Decision Tree.

- **Root Decision Node**: *General Health* ≤ 2.5. The model isolates subjective health rating as the single most discriminative split. Patients with Fair/Poor health immediately route to high-risk branches.
- **Secondary Metabolic Splits**: *High Blood Pressure* (≤ 0.5) further stratifies risk across both wellness categories.
- **Tertiary Interaction Nodes**: Specific metabolic cutoffs emerge naturally: BMI > 30.5 (clinical Class I obesity threshold) and Age > 51.5.
- **Clinical Auditability (‘trace_path‘)**: The decision path of any individual patient can be extracted in constant time as human-readable if-then rules (e.g., General Health $> 2.5 \wedge$ Hypertension $= 1 \wedge$ Age $> 55.5 \implies$ High Risk), fostering clinician trust and enabling deployment in offline triage checklists.

8 Conclusion & Student Reflection

Through this project, we demonstrated that raw algorithmic capacity does not equate to clinical utility. Unrestricted decision trees severely overfit survey noise (34% generalization gap). Applying early stopping pre-pruning and the **1% Tolerance Rule** enabled us to build a compact 63-node tree that captures 77.7% of prediabetes/diabetes cases with 72.56% test accuracy. Ensuring strict leakage-free imputation and evaluating Macro F_1 guarantees that the resulting model is mathematically sound, clinically aligned, and transparent for real-world deployment.